— PROJECT PI² · ESILV ENERGY & SUBSTAINABLE CITIES

Min Cost Flow Simulator

Physics Optimization & Energy Markets

Project 196

Team 4326

1 Context & Challenges

The energy transition demands complex grid management to integrate **intermittent renewables** and optimize costs at national scale. This simulator models France's grid to address three core challenges:

- **Balancing:** Match supply and demand in real-time.
- **Finance:** Minimize production costs via the SPOT market (**Merit Order**).
- **Security:** Prevent blackouts and manage congestion (**N-1 Crash Test**).

3 Modeling & Resolution

France's grid is abstracted as a **directed graph G = (N, A)** via **NetworkX**. This is a **Minimum Cost Flow** optimization problem: nodes are cities or plants; arcs are high-voltage lines with capacity & cost constraints.

MIN-COST FLOW ALGORITHM

Busacker–Gowen

Successive Shortest Paths method — iteratively solves the economic dispatch problem step by step.

01 **Search** for the cheapest augmenting path (Bellman-Ford).

02 **Saturate** the line along this path up to its capacity.

03 **Update** the residual graph and repeat until equilibrium.

★ **Carbon Tax** integrated dynamically — modifies arc costs live to reflect CO_2 pricing in the dispatch.

2 Software Architecture

Data stored in structured **CSV files**: power plants, cities, and high-voltage transmission lines.

⚙ Backend · Python

FastAPI | NetworkX | Pandas

Computation engine for **Min-Cost Flow** physical optimization & API exposure.

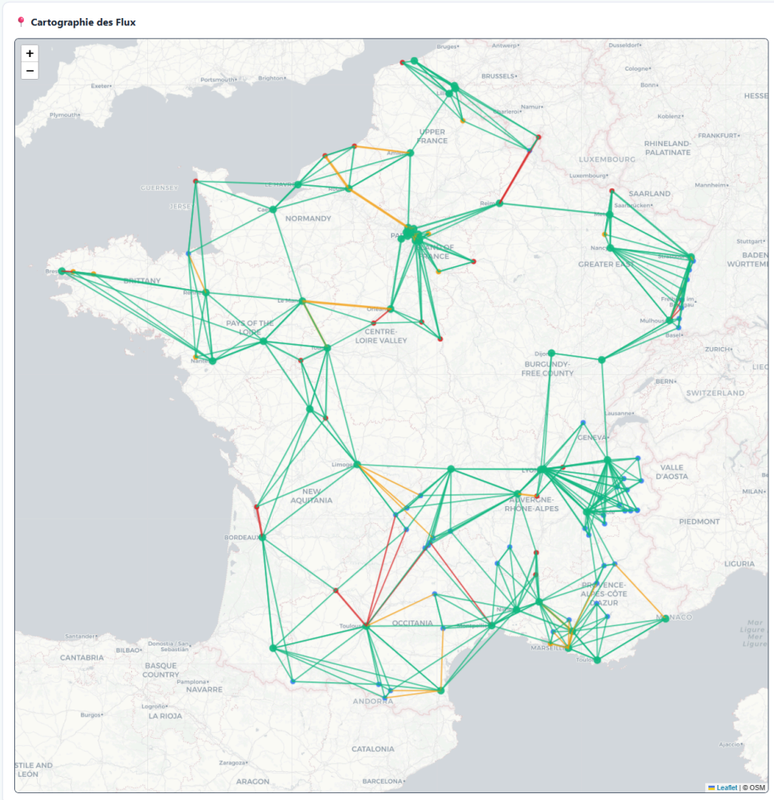
□ Frontend · JavaScript

Leaflet.js | Tailwind | Chart.js

Real-time SCADA dashboard with **geocoded map** and dynamic energy charts.

4 SCADA Dashboard & Crash Test

The dashboard simulates the full **24h daily load cycle** (Unit Commitment) hour by hour — showing the map of France with line congestion states (green / orange / red), the stacked **energy mix**, and the **N-1 Crash Test** resilience analysis.



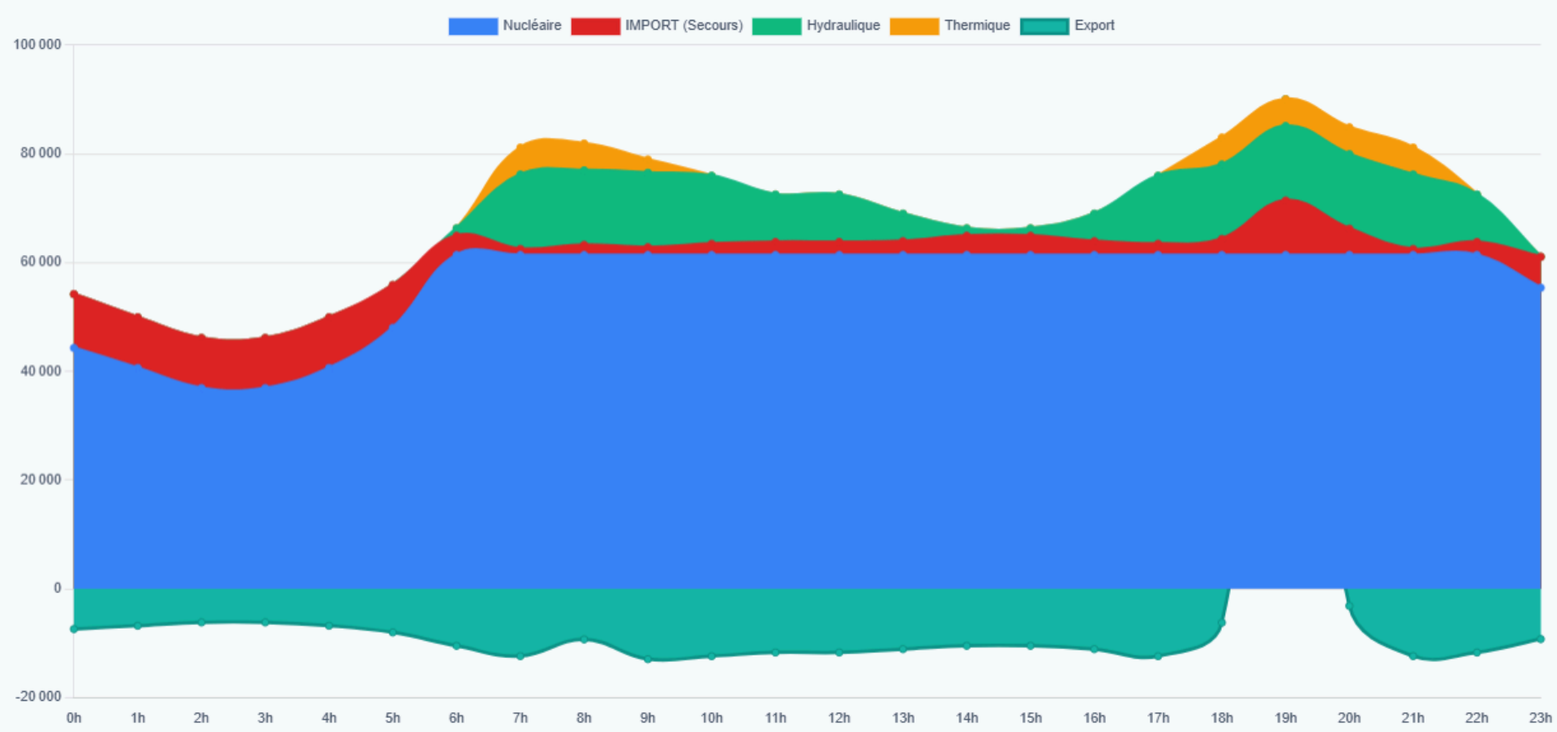


Figure 1. Top: Real-time geocoded map & line saturation. Bottom: Energy mix stack (24h).

5 Results & Prototype Contributions

- **Algorithm covers 100% of demand** while respecting all physical line constraints.
- **Full dynamic graph** built from CSV — automatic geocoding of all nodes.
- **Optimal flows** computed on every arc; total operational cost minimized in real-time.

100%
DEMAND COVERAGE

< 1s
COMPUTATION TIME